

COLD HEADER - Line A (4.5 K) / Line B (2 K)

THERMAL SHIELD - Line D (40 K in) / Line E (60 K out)

EQUIPMENT cavities / vessels / heat exchangers

WARM PIPING SYSTEM (WPS) - Line W / S / U -> QRB handover

LINE W - TEMPERATURE GRADIENT (WPS warm return)

W @ 4.5 K ambient / electrical heater QCELL side (cold) warming USER side -> QRB handover W @ 300 K

INTERACTIVE LEGEND

toggle: layer "3d_Legend_INTERACTIVE"

LINE SPECIFICATION TABLE

LINE	TEMP	PRESS	DN	MOC
COLD HEADER				
A	4.5 K	3 bar	DN50	SS316L
A'	4.5 K	3 bar	DN25	SS316L
B	2 K	27 mbar	DN40	SS316L
B'	2 K	27 mbar	DN25	SS316L
THERMAL SHIELD				
D	40 K	14 bar	DN32	CU
D'	40 K	14 bar	DN20	CU
E	60 K	13 bar	DN32	CU
E'	60 K	13 bar	DN20	CU
WARM (WPS)				
W	4.5 K-300	6 bar	DN20	SS304
S	2-292 K	1.05 bar	DN15	SS304
U	292 K	14 bar	DN15	SS304

REFERENCE

Material colour -> line weight + dash + on-line NAME/LABEL

SIGNAL TYPES (0.25 mm)

- Pneumatic (dash + //)
- Electric (dotted)
- Hydraulic (dash-dot)

INSTRUMENTS / SCOPE

- Field instrument (white box tag)
- Panel / safety (line)
- TP scope diamond (edge = TP layer)

Cat: B=Building, C=Chil, E=Electrical, G=Compressed gases

HH=HVAC, LL=Liquid waste, S=Solid waste, W=Water

CONSULTANT Mott MacDonald Bristol, UK	CLIENT SCK CEN Boerstang 200, 2400 Mxl, BE MYRRHA / MINERVA Phase 1	REVISIONS Rev Date Description C1 2026-06 v3 refinement - layers/egbil B1 2026-05 v2 split sheets A1 2026-04 First issue	App'd ACR ACR ACR	APPROVALS Designed QSYS Drawn ACR Checked ACR Approved PM	MINERVA CryoCell - SCK CEN RFCELL (seen by ACR) - Process Flow (DI-Wa DWG No. HA-P581_PP8713 MMD Proj. 411266 Scale NTS Suitability S2 - FOR ACCEPTANCE Security RESTRICTED Size / Std A3 / ISO1628 / ISA-S.1 Sheet Sheet1 of 2 - Process Variant CONTROL-CENTRIC
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